

1 Gain law

(bars denote normalized quantities: lengths by R , $\bar{a} = a/a_o$, $\bar{f} = a_o f$; $0 < \bar{a} < 1$; $e_\theta = \theta - \hat{\theta}$)

$$\bar{a}_{\text{true}}(\bar{w}) = 1 - \frac{9}{8\bar{w}} + \frac{1}{2\bar{w}^3} - \frac{57}{100\bar{w}^4} + \frac{1}{5\bar{w}^5} + \frac{7}{200\bar{w}^{11}} - \frac{1}{25\bar{w}^{12}}$$

$$1 - \bar{a}_{\text{true}} = \frac{9}{8\bar{w}} + \mathcal{O}(\bar{w}^{-3}), \quad \bar{a}'_{\text{true}} = \frac{9}{8\bar{w}^2} + \mathcal{O}(\bar{w}^{-4})$$

$$\bar{a}'(\bar{a}, b) := \frac{d\bar{a}}{d\bar{w}} = \frac{(1 - \bar{a})^2}{b}, \quad a_w = a_o \bar{a}_w, \quad a_o = \frac{\Delta t}{\gamma_N R}$$

$$b_{\text{true}}(\bar{w}) := \frac{(1 - \bar{a}_{\text{true}})^2}{\bar{a}'_{\text{true}}} = \frac{9}{8} + \frac{1}{2\bar{w}^2} - \frac{57}{50\bar{w}^3} + \mathcal{O}(\bar{w}^{-4})$$

$$\bar{a}(\bar{w}) = 1 - \frac{b}{\bar{w} - \bar{w}_0} \quad (\text{integrated law; used at the seed only})$$

2 Jacobians

$$\frac{\partial \bar{a}'}{\partial \bar{a}} = -\frac{2(1 - \bar{a})}{b}, \quad \frac{\partial \bar{a}'}{\partial b} = -\frac{(1 - \bar{a})^2}{b^2} = -\frac{\bar{a}'}{b}$$

$$\left. \frac{\partial \bar{a}}{\partial b} \right|_{\bar{w}} = -\frac{1}{\bar{w} - \bar{w}_0} = -\frac{1 - \bar{a}}{b}, \quad \left. \frac{\partial \bar{a}}{\partial \bar{w}_0} \right|_{\bar{w}} = -\bar{a}'$$

3 True dynamics

$$(\delta \bar{w}_j[k] := \delta \bar{w}[k - d + j - 1], \quad j = 1, \dots, d + 1, \quad d = 2)$$

$$\begin{aligned} \bar{w}[k] &= \bar{w}_d[k] - \delta \bar{w}_3[k], & \bar{w}_d[k + 1] &= \bar{w}_d[k] + \Delta \bar{w}_d[k] \\ \delta \bar{w}_1[k + 1] &= \delta \bar{w}_2[k], & \delta \bar{w}_2[k + 1] &= \delta \bar{w}_3[k] \end{aligned}$$

$$F_{dw}[k] = \bar{f}_{dw}[k] + (1 - \lambda_c) \sum_{i=1}^d \bar{f}_{dw}[k - i]$$

$$\varepsilon_w[k] = \bar{a}_w[k] \bar{f}_T[k] + (1 - \lambda_c) \sum_{i=1}^d \bar{a}_w[k - i] \bar{f}_T[k - i] + (1 - \lambda_c) n_w[k]$$

(a)

$$\delta \bar{w}_3[k+1] = \lambda_c \delta \bar{w}_3[k] - F_{dw}[k] e_{\bar{a}_w}[k] - \varepsilon_w[k]$$

$$\bar{a}_w[k+1] = \bar{a}_w[k] + \bar{a}'_{\text{true}}[k] \left[\Delta \bar{w}_d[k] + (1 - \lambda_c) \delta \bar{w}_3[k] + F_{dw}[k] e_{\bar{a}_w}[k] + \varepsilon_w[k] \right], \quad b[k+1] = b[k]$$

(b)

$$\bar{w}_D[k] := \bar{w}_{D,0}[k] + (1 - \lambda_c) \sum_{i=1}^d \bar{w}_{D,0}[k-i]$$

$$\delta \bar{w}_3[k+1] = \lambda_c \delta \bar{w}_3[k] - F_{dw}[k] e_{\bar{a}_w}[k] - e_{\bar{w}_D}[k] - \varepsilon_w[k]$$

$$\bar{a}_w[k+1] = \bar{a}_w[k] + \bar{a}'_{\text{true}}[k] \left[\Delta \bar{w}_d[k] + (1 - \lambda_c) \delta \bar{w}_3[k] + F_{dw}[k] e_{\bar{a}_w}[k] + e_{\bar{w}_D}[k] + \varepsilon_w[k] \right]$$

$$b[k+1] = b[k], \quad \bar{w}_D[k+1] = \bar{w}_D[k] + \delta \bar{w}_D[k], \quad \delta \bar{w}_D[k+1] = \delta \bar{w}_D[k]$$

4 Process model

$$(a) \quad \mathbf{x} = [\delta \bar{w}_1, \delta \bar{w}_2, \delta \bar{w}_3, \bar{a}_w, b]^\top$$

$$(b) \quad \mathbf{x} = [\delta \bar{w}_1, \delta \bar{w}_2, \delta \bar{w}_3, \bar{a}_w, b, \bar{w}_D, \delta \bar{w}_D]^\top \quad (d=2)$$

5 Estimator

$$\delta \hat{w}_1[k+1] = \delta \hat{w}_2[k], \quad \delta \hat{w}_2[k+1] = \delta \hat{w}_3[k], \quad \delta \hat{w}_3[k+1] = \lambda_c \delta \hat{w}_3[k]$$

$$\hat{a}_w[k+1] = \hat{a}_w[k] + \frac{(1 - \hat{a}_w[k])^2}{\hat{b}[k]} \left[\Delta \bar{w}_d[k] + (1 - \lambda_c) \delta \hat{w}_3[k] \right], \quad \hat{b}[k+1] = \hat{b}[k]$$

$$(b) \quad \hat{w}_D[k+1] = \hat{w}_D[k] + \delta \hat{w}_D[k], \quad \delta \hat{w}_D[k+1] = \delta \hat{w}_D[k]$$

6 Error dynamics

(True – Estimator; coefficients at the estimate)

$$\begin{aligned}
e_{\delta\bar{w}_1}[k+1] &= e_{\delta\bar{w}_2}[k], & e_{\delta\bar{w}_2}[k+1] &= e_{\delta\bar{w}_3}[k] \\
e_{\delta\bar{w}_3}[k+1] &= \lambda_c e_{\delta\bar{w}_3}[k] - F_{dw}[k] e_{\bar{a}_w}[k] - \varepsilon_w[k] \quad [-e_{\bar{w}_D}[k] \text{ in (b)}] \\
e_{\bar{a}_w}[k+1] &= \left[1 + \frac{(1 - \hat{a}_w)^2}{\hat{b}} F_{dw} - \frac{2(1 - \hat{a}_w)}{\hat{b}} (\Delta\bar{w}_d + (1 - \lambda_c)\delta\hat{w}_3) \right] e_{\bar{a}_w}[k] \\
&\quad + \frac{(1 - \lambda_c)(1 - \hat{a}_w)^2}{\hat{b}} e_{\delta\bar{w}_3}[k] - \frac{(1 - \hat{a}_w)^2}{\hat{b}^2} (\Delta\bar{w}_d + (1 - \lambda_c)\delta\hat{w}_3) e_b[k] \\
&\quad + \frac{(1 - \hat{a}_w)^2}{\hat{b}} \varepsilon_w[k] \quad \left[+ \frac{(1 - \hat{a}_w)^2}{\hat{b}} e_{\bar{w}_D}[k] \text{ in (b)} \right] \\
e_b[k+1] &= e_b[k] \\
e_{\bar{w}_D}[k+1] &= e_{\bar{w}_D}[k] + e_{\delta\bar{w}_D}[k], & e_{\delta\bar{w}_D}[k+1] &= e_{\delta\bar{w}_D}[k]
\end{aligned}$$

$$\text{(b)} \quad F_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & \lambda_c & -F_{dw} & 0 & 0 \\ 0 & 0 & \frac{(1 - \lambda_c)(1 - \hat{a}_w)^2}{\hat{b}} & 1 + \frac{(1 - \hat{a}_w)^2}{\hat{b}} F_{dw} - \frac{2(1 - \hat{a}_w)}{\hat{b}} (\Delta\bar{w}_d + (1 - \lambda_c)\delta\hat{w}_3) & -\frac{(1 - \hat{a}_w)^2}{\hat{b}^2} (\Delta\bar{w}_d + (1 - \lambda_c)\delta\hat{w}_3) & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

(a) is the leading 5×5 block.

$$\mathbf{q}[k] = \left[0, 0, -\varepsilon_w, \frac{(1 - \hat{a}_w)^2}{\hat{b}} \varepsilon_w, 0, 0, 0 \right]^\top$$

$$e_{\bar{a}}(\bar{w})|_{\bar{w}_0} = -\bar{a}'(\bar{w}) e_{\bar{w}_0} \propto \frac{b}{(\bar{w} - \bar{w}_0)^2}, \quad e_{\bar{a}}(\bar{w})|_b = -\frac{e_b}{\bar{w} - \bar{w}_0}$$

7 Measurement and innovation

$$\begin{aligned}
\nabla_d \bar{w}_d[k] &= \bar{w}_d[k] - \bar{w}_d[k-d] \\
y_1[k] &= \delta\bar{w}_1[k] + n_w[k] \\
y_2[k] &= \bar{a}_w[k-d] + n_a[k]
\end{aligned}$$

$$\bar{a}_w[k-d] = \bar{a}_w[k] - \frac{(1 - \bar{a}_w[k])^2}{b} \nabla_d \bar{w}_d[k]$$

$$\frac{\partial y_2}{\partial \bar{a}_w} = 1 + \frac{2(1 - \hat{a}_w)}{\hat{b}} \nabla_d \bar{w}_d, \quad \frac{\partial y_2}{\partial b} = \frac{(1 - \hat{a}_w)^2}{\hat{b}^2} \nabla_d \bar{w}_d$$

$$(b) \quad H[k] = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 + \frac{2(1 - \hat{a}_w)}{\hat{b}} \nabla_d \bar{w}_d & \frac{(1 - \hat{a}_w)^2}{\hat{b}^2} \nabla_d \bar{w}_d & 0 & 0 \end{bmatrix}, \quad K[k] = P^f H^\top (H P^f H^\top + R)^{-1}$$

$$e_{y_1}[k] = e_{\delta \bar{w}_1}[k] + n_w[k]$$

$$e_{y_2}[k] = \left[1 + \frac{2(1 - \hat{a}_w)}{\hat{b}} \nabla_d \bar{w}_d \right] e_{\bar{a}_w}[k] + \frac{(1 - \hat{a}_w)^2}{\hat{b}^2} \nabla_d \bar{w}_d e_b[k] + n_a[k]$$

8 Process and measurement noise

$$q_3 = -\bar{a}_w \bar{f}_T, \quad q_4 = -\frac{(1 - \hat{a}_w)^2}{\hat{b}} q_3, \quad \gamma(\bar{w}) = \gamma_{N^c}(\bar{w}), \quad \sigma_{\bar{f}_T}^2 = a_o^2 \frac{4k_B T \gamma(\bar{w})}{\Delta t}$$

$$Q_{33} = \frac{4k_B T a_o}{R} \bar{a}_w, \quad Q_{34} = Q_{43} = -\frac{(1 - \hat{a}_w)^2}{\hat{b}} Q_{33}, \quad Q_{44} = \frac{(1 - \hat{a}_w)^4}{\hat{b}^2} Q_{33}$$

$$Q_{55} = 0, \quad Q_{66} = Q_{77} = 0$$

$$R = \text{diag}(R_1, R_2), \quad R_1 = \sigma_{n_w}^2, \quad R_2 = K_{\text{var}} I F_{\text{var}} (\bar{a}_w + \xi)^2 + d Q_{44}, \quad K_{\text{var}} = \frac{2a_{\text{cov}}}{2 - a_{\text{cov}}}$$

$$\sigma_{\delta \bar{w}_r}^2 = C_{dpmr} \frac{4k_B T a_o}{R} (\bar{a}_w + \xi), \quad \xi = \frac{C_n \sigma_{n_w}^2}{C_{dpmr} (4k_B T a_o / R)}$$

Seed and prior

$$\hat{b}[0] = \frac{9}{8} = \lim_{\bar{w} \rightarrow \infty} b_{\text{true}}(\bar{w}), \quad \hat{a}_w[0] = 1 - \frac{\hat{b}[0]}{\bar{w}[0] - \hat{w}_0}, \quad \hat{w}_D[0] = \delta \hat{w}_D[0] = 0$$

$$P_{55}[0] = P_{bb} = \left[\sup_{\bar{w} \in \mathcal{E}} \left| b_{\text{true}}(\bar{w}) - \frac{9}{8} \right| \right]^2$$

$$P_{44}[0] = (\hat{a}'[0])^2 P_{\bar{w}_0} + \left(\frac{1 - \hat{a}_w[0]}{\hat{b}[0]} \right)^2 P_{bb} + \text{floor}_a^2, \quad P_{45}[0] = - \frac{1 - \hat{a}_w[0]}{\hat{b}[0]} P_{bb}$$

$$P_{66}[0] = P_D, \quad P_{77}[0] = P_{\delta D}, \quad P_{46}[0] = P_{47}[0] = P_{56}[0] = P_{57}[0] = P_{67}[0] = 0$$